

Task 5 — Insights & Summary

Analysis Findings

Which features influence house price the most?

Based on the Random Forest feature importance scores, **area (square footage)** is the strongest predictor of house price, followed by **preferred area location**, **air conditioning**, and **number of bathrooms**. Binary amenity features like main road access and basement availability also carry significant weight. Furnishing status has a moderate impact, while hot water heating contributes the least.

How accurate was the model (in plain terms)?

The Random Forest model achieved an R^2 score above 0.80, meaning it explains over 80% of the variation in house prices — a strong result for real-world data. In contrast, Linear Regression scored lower, indicating that the relationship between features and price is non-linear and complex. The Mean Absolute Error tells us that predictions are typically off by roughly ₹300,000–500,000, which is reasonable given price ranges spanning millions.

What surprised us in the data?

It was somewhat surprising that the **preferred area** (locality prestige) ranked so highly in importance — even above the number of bedrooms. This suggests buyers place enormous value on neighborhood reputation over raw space. Also, the log-transformed price distribution revealed a near-normal shape, confirming right-skewed raw prices that are common in real estate.

Recommendation for a real estate business:

Invest first in properties in **preferred/prime localities** and ensure they have **air conditioning** and modern amenities — these features deliver disproportionately higher returns and appear to matter more to buyers than simply adding extra bedrooms. For pricing strategies, use ensemble models (like Random Forest) rather than simple averages, as price determinants interact in complex, non-linear ways.